

Objectives

- ▶ Study whether **hard negatives** (plausible but wrong training examples) improve link prediction in knowledge graphs.
- ▶ Design a **two-stage sampling pipeline** that upgrades negative quality at *zero extra computational cost*.
- ▶ Compare five strategies (Random, Hard, Mixed 30/50/70 %) on RotatE / FB15k-237 under identical hyperparameters.

Problem

A **knowledge graph** (KG) is a network of facts stored as triples (**subject, relation, object**). **Link prediction** asks: given $(h, r, ?)$, which entity t is most likely to complete the triple?



Models learn by comparing correct triples against *negatives* (fabricated wrong triples). The quality of negatives is critical.

RANDOM NEGATIVES (STANDARD)

Obama → born.in → *Ice cream*
Obama → born.in → *Mt. Everest*
Obama → born.in → *Pacific Ocean*

Too easy. Model stops improving.

HARD NEGATIVES (OURS)

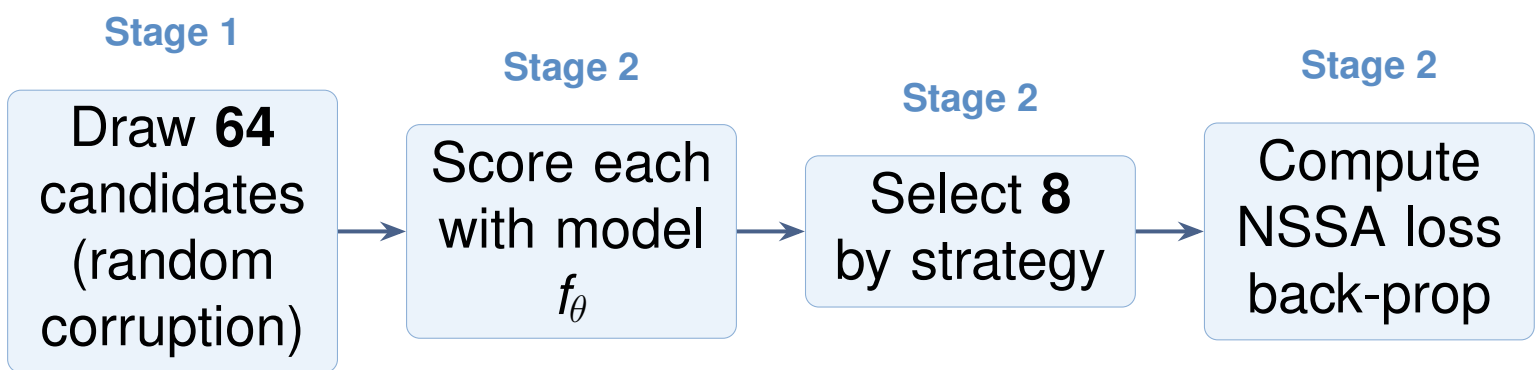
Obama → born.in → *Chicago*
Obama → born.in → *Kansas City*
Obama → born.in → *Honolulu*

Plausible but wrong. Model keeps improving.

Honolulu is the correct answer here, making it the hardest possible negative for this query.

Methodology

For each positive triple at training time:



Selection strategies tested in Stage 2:

- **Random**: sample 8 uniformly (control)
- **Hard**: take the 8 highest-scoring wrong candidates
- **Mixed p** : take fraction p hard + $(1 - p)$ random, with $p \in \{30\%, 50\%, 70\%\}$

All strategies share the same Stage 1 pool: zero extra compute.

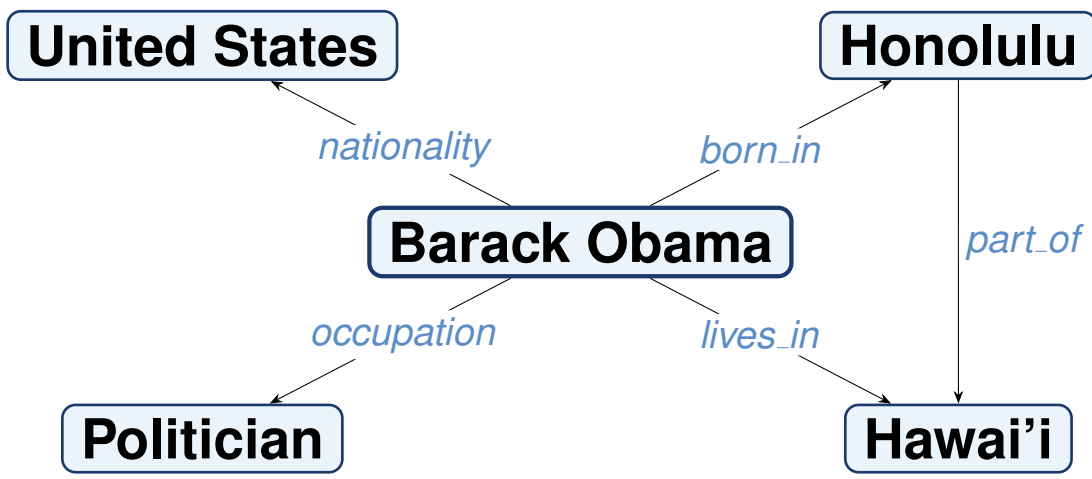
Common setup: RotatE, embedding dim 128, batch 1024, Adam lr= 10^{-3} , 50 epochs (patience 10), seed 42.

References

- [1] Sun *et al.* (2019). *RotatE: Knowledge Graph Embedding by Relational Rotation in Complex Space*. ICLR 2019.
- [2] Cai & Wang (2018). *KBGAN: Adversarial Learning for Knowledge Graph Embeddings*. NAACL 2018.
- [3] Toutanova & Chen (2015). *Observed versus latent features for KB and text inference*. ACL Workshop 2015 (FB15k-237 dataset).

Datasets

FB15k-237 is a widely used benchmark knowledge graph derived from Freebase, covering people, places, films, and organisations. Each node is an **entity**; each labelled edge is a **fact**.



Entities: 14,505 Train triples: 272,115
Relation types: 237 Val / Test: 17,526 / 20,438
Relation frequencies range over **three orders of magnitude** (37 to 15,989 occurrences), enabling stratified evaluation by rarity.

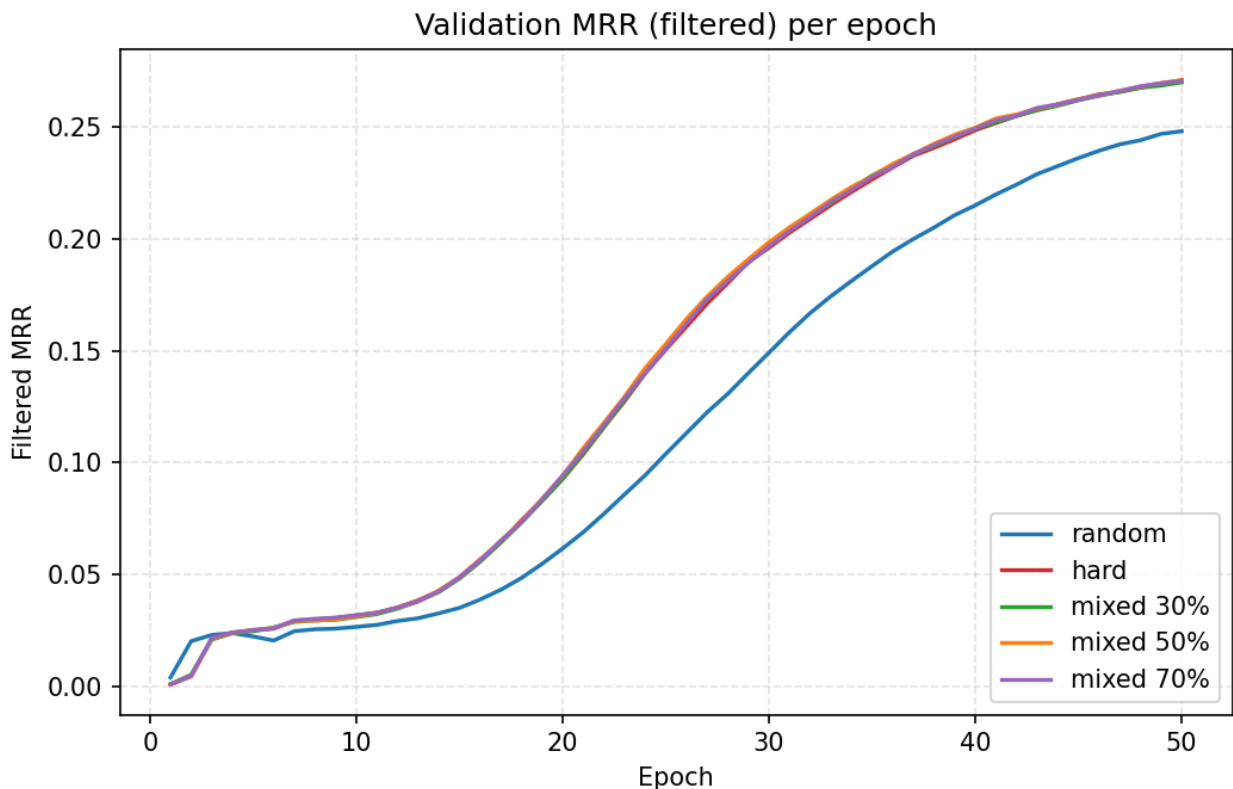
Results

+10.4%

accuracy gain (MRR) with Hard negatives
vs. standard random sampling

Strategy	MRR	H@1	H@3	H@10
Random (baseline)	0.270	0.189	0.300	0.428
Hard	0.298	0.217	0.327	0.459
Mixed 30%	0.297	0.215	0.326	0.459
Mixed 50%	0.298	0.216	0.328	0.460
Mixed 70%	0.299	0.217	0.328	0.461
Δ Hard – Random	+0.028	+0.028	+0.026	+0.031

Mixed 30% hard recovers 95% of the gain at zero extra cost.



Hard and mixed strategies **pull ahead from random after epoch 15** and converge to a 7–8 % higher plateau with no training instability.

Stratified evaluation, MRR gain Hard vs. Random:

+3.6 pp Rare relations 0.382 → 0.418	+3.0 pp Medium relations 0.332 → 0.362	+2.8 pp Frequent relations 0.257 → 0.285
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Hard negatives benefit every slice, but gain is largest for **rare relations** where random negatives are least informative.

Conclusions

- ▶ **Hard negatives are a free upgrade**: same training pipeline, smarter selection step, +10.4% MRR gain.
- ▶ **You don't need 100% hard**: Mixed 30% (2 hard examples out of 8) recovers 95% of the gain with added stability.
- ▶ **Rare and ambiguous facts benefit most**, which is exactly where KG completion is hardest and most valuable in practice.
- ▶ **Future work**: dynamically adjust the hard fraction during training; scale to Wikidata-5M; test on RESCAL and TuckER.